

Customer Transaction Analysis — Nigerian FinTech Platform

How the Excel analysis workbook was built, checked and delivered · 28 September 2026

1. Summary

A 2,500-row transaction dataset for a Nigerian FinTech platform was turned into an executive-standard Excel workbook with **nine sheets** (eight visible and one hidden helper sheet). It answers six management objectives and ends in an interactive C-suite dashboard.

Every analytical figure is a live Excel formula linked to the source table. The dashboard is driven by PivotTables and filtered by three connected slicers. The final file contains **2,218 formulas with zero errors**, and its headline figures were independently confirmed in Python.

Item	Detail
Deliverable	FinTech_Transaction_Analysis_Executive_Workbook.xlsx
Source file	FinTech_Customer_Transaction_Analysis11.xlsx (left unchanged)
Tools used	Python (pandas, openpyxl), Microsoft Excel automation (COM / pywin32)
Status	Complete: verified for formula errors, figure accuracy and slicer behaviour

2. Source Data

The dataset was profiled before any building started, to confirm it was complete and to learn its exact category labels.

Attribute	Finding
Records	2,500 transactions, 11 columns
Period	1 January 2025 – 30 December 2025
Data quality	No missing values and no duplicate Transaction IDs
Customers	350 unique customers; each has one State and one Account Type
States	10: Lagos, Kano, Oyo, FCT Abuja, Rivers, Delta, Ogun, Edo, Enugu, Kaduna
Transaction types	Transfer, Airtime/Data, Bill Payment, POS Payment, Card Withdrawal
Account types	Savings, Current, Tier 1 Wallet, Tier 3 Verified
Payment methods	Bank Transfer, Debit Card, USSD, Wallet Balance

3. How the Task Was Carried Out

The work followed five stages. Building in two stages played to each tool's strengths: Python for bulk data, formulas and formatting, and Excel itself for native charts, PivotTables and slicers, which Python libraries cannot create reliably.

Stage 1 — Data inspection and profiling

- Located the dataset in the project folder and read it with pandas.
- Checked data types, missing values, duplicates, date range and category labels.
- Pre-computed every key metric in Python. These figures later served as an independent benchmark for checking the Excel formulas.

Stage 2 — Workbook build with formulas (Python + openpyxl)

- **Raw_Data:** wrote the 2,500 rows as an Excel Table named TransactionTable, with ₦#,##0.00 amounts and YYYY-MM-DD HH:MM dates.
- **Analysis sheets (Obj1–Obj5):** built every table with live formulas (SUM, COUNTA, AVERAGE, COUNTIF(S), SUMIF(S), AVERAGEIF) that reference TransactionTable by column name.
- **Self-sorting rankings:** the Top 10 customers and the state ranking use LARGE + INDEX/MATCH over supporting ledgers, so they re-sort automatically when the data changes.
- **Executive brief (Obj6):** wrote the management answers as formula-built sentences, so the numbers inside the text update with the data.
- **Design system:** applied the Corporate Navy & Soft Blue palette, Calibri font, and consistent ₦ / 0.0% / #,##0 number formats throughout.

Stage 3 — Formula validation in Excel

- Opened the stage-one file in Microsoft Excel, forced a full recalculation, and scanned every formula cell for errors.
- Compared key results against the Python benchmark; all matched exactly.

Stage 4 — Charts, PivotTables, slicers and dashboard (Excel automation)

- Drove Excel directly from Python to add native charts to each analysis sheet: a donut, horizontal bars, a dual-axis clustered column and a column + line combination.
- Created one shared PivotCache and six PivotTables (KPIs, status, month, state, product, Top 10) on a hidden Pivot_Engine sheet and the dashboard.
- Added three slicers (State, Account_Type, Payment_Method) and connected each one to all six PivotTables.
- Built the dashboard: five KPI cards linked to GETPIVOTDATA formulas, four pivot charts and a styled Top 10 PivotTable.

Stage 5 — Quality assurance

- Exported each sheet to PDF and reviewed the layout, charts and labels visually; fixed any issues found.
- Tested the slicers by filtering to Lagos and confirming that the dashboard KPIs matched the Python figures for Lagos.
- Re-scanned the final workbook: 2,218 formulas, 0 errors.

4. Workbook Structure

Sheet	Contents
1. Raw_Data	TransactionTable: all 2,500 transactions, formatted and frozen header
2. Obj1_Volume_and_Value	KPI summary with visible formulas, operational health table, reconciliation check, donut chart
3. Obj2_High_Value_Customers	Top 10 customers, account-type and age-group analysis, 350-customer ledger, horizontal bar chart
4. Obj3_Product_Popularity	Product matrix, payment channel table, volume vs value ranks, dual-axis column chart
5. Obj4_Geographic_Distribution	Ranked state table, key-hub highlighting and concentration table, bar chart
6. Obj5_Growth_Trajectory	12-month trend with MoM growth, growth diagnostics, column + line combo chart
7. Obj6_Business_Drivers_Summary	Executive summary, answers to management questions, four driver commentaries with actions
Pivot_Engine (hidden)	Shared PivotTables and GETPIVOTDATA KPI block feeding the dashboard
8. Executive_Dashboard	5 KPI cards, 4 pivot charts, Top 10 table and 3 connected slicers; gridlines hidden

5. Key Findings

- **Scale:** ₦36,963,242.61 transacted across 2,500 transactions (average ₦14,785.30).
- **Reliability:** 88.3% successful, 9.6% failed, 2.1% pending. Failed transactions account for ₦3.74M, or 10.1% of attempted value.
- **Geography:** Lagos generates 38.6% of value. Lagos, FCT Abuja and Rivers together generate 55.4%.
- **Products:** Transfers are 43.4% of transactions but 75.4% of value. Airtime/Data is 26.4% of transactions but only 3.1% of value.
- **Customers:** Revenue is broad-based: the Top 10 customers hold only 7.8% of value. Savings accounts (50.0%) and the 35–49 age group (43.4%) lead.
- **Growth:** Monthly value is volatile (–34.1% to +58.4% month on month), but the year trends upward: H2 value was 5.0% above H1.

6. Quality Assurance Results

Check	Result
Formula error scan (all sheets)	2,218 formulas, 0 errors
Excel totals vs Python benchmark	All headline figures match exactly
Status table reconciliation	Counts and values reconcile to KPI totals
Slicer test (State = Lagos)	985 transactions, ₦14,472.47 average (matches Python)
Slicer connections	Each slicer connected to all 6 PivotTables
Visual review	All sheets exported to PDF and checked

7. Issues Encountered and How They Were Resolved

Issue	Resolution
FORMULATEXT returned #NAME? errors	Added the _xlfn. prefix that Excel requires for newer functions in saved files
Excel automation calls rejected while Excel was busy	Added an automatic retry wrapper around every Excel call
Leftover background Excel processes locked the file	Closed the orphaned instances and added guaranteed clean-up
Cell edits refused while a chart was selected	Returned the selection to a cell before each group of edits
Dashboard Success/Failure rates were swapped	Corrected an off-by-one row reference in the KPI engine
Slicers stacked over the title banner	Set each slicer's position explicitly after creation
State chart labels overlapped state names	Moved labels outside the bars and widened the axis

8. Design Decisions and Deviations from the Brief

- **Hidden Pivot_Engine sheet:** slicers can only filter PivotTables, not ordinary formulas, so the dashboard's pivots live on a hidden ninth sheet.
- **Chart axes:** use a compact ₦ millions format (e.g. ₦4.0M) for readability. All cells and data labels use ₦#,##0.00.
- **Category labels:** the source's own names were kept ("POS Payment", "Tier 1 Wallet", "Tier 3 Verified").
- **Revenue definition:** "Revenue" and "Total Spent" include all statuses, consistent with Total Transaction Value. Successful-only value (₦32.3M) is reported separately in Obj1.
- **Age bands:** <25, 25–34, 35–49 and 50+ are stored as editable input cells, so they can be re-banded.

9. How to Use and Update the Workbook

1. Open the workbook; it opens on the Executive_Dashboard.
2. Click slicer buttons to filter by State, Account Type or Payment Method. Use the X at the top-right of a slicer to clear it.
3. To add data, paste new rows directly beneath TransactionTable on Raw_Data; the table expands automatically.
4. Select Data → Refresh All so the dashboard PivotTables pick up the new rows. The analysis sheets recalculate on their own.